

AI-based Super-Resolution Diffusion MRI Unveils Heterogeneous Cortical Microstructural Alterations in Alzheimer's Disease

Xiaoxu Zheng¹, Haotian Li¹, Zhibo Chen¹, Yongquan Huang¹, Jue Lu¹, Ruike Chen¹,
Ziyu He¹, Sihan Ding¹, Dan Wu^{1*}

¹Department of Biomedical Engineering, College of Biomedical Engineering &
Instrument Science, Zhejiang University, Hangzhou, China

*Corresponding authors: Dr. Wu Dan, e-mail: danwu.bme@zju.edu.cn

Introduction

Identifying reliable biomarkers for Alzheimer's disease (AD) is critical, and while the amyloid-tau-neurodegeneration (A/T/N) framework provides a robust biological definition of the disease, current non-invasive assessments of neurodegeneration (N) are often constrained by the limited sensitivity of standard structural MRI. Conventional volumetric markers typically captured macroscopic atrophy, which was a relatively late downstream event and manifested only after substantial neuronal loss has occurred¹. Therefore, it was still necessary to detect the initial microstructural breakdown before any morphological changes occur. However, while diffusion MRI (dMRI) could probe microstructural integrity, standard clinical acquisitions, such as the Alzheimer's Disease Neuroimaging Initiative (ADNI), lacked the resolution to resolve fine-scale cortical columns, forcing studies to rely on "averaged" metrics that hid laminar-specific heterogeneity.

To overcome this, we modified the Disentangled Conditional Diffusion model (DisC-Diff)², an advanced generative AI, which super-resolved clinical dMRI to submillimeter precision while preserving high-frequency anatomical details. Leveraging this enhanced resolution, we focused on the Radiality Index (RI)³, a metric specifically designed to quantify the coherence of water diffusion along the vertical axis of cortical columns. Due the disruption of these vertical mini-columns (loss of dendritic bundles) was a biomarker of early AD pathology, RI served as a sensitive in vivo proxy for column integrity. Applying this framework to the ADNI cohort to sample laminar RI profiles, we revealed spatial heterogeneity in AD microstructural breakdown, characterized by two distinct signatures: a laminar profile asymmetry (high RI Skewness) predominating in posterior networks, and an attenuated peak radial integrity (low RI Maximum) concentrated in frontal regions.

Method

MRI Acquisition and Image Preprocessing

Data were obtained from the ADNI database⁴ (adni.loni.usc.edu). We selected participants who scanned both high-resolution 3D T1-weighted structural MRI and diffusion-weighted imaging (DWI) on 3T MRI scanners. The standard ADNI diffusion protocol consisted of a voxel size of approximately $2 \times 2 \times 2 \text{ mm}^3$, with 48 diffusion-encoding directions at a b-value of 1000 s/mm^2 and six b0 images. Corresponding T1-weighted images were acquired using an MPRAGE sequence at $1 \times 1 \times 1 \text{ mm}^3$ resolution. The final study cohort included 156 participants, categorized into Healthy Controls (HC), Mild Cognitive Impairment (MCI), and AD groups.

The T1-weighted structural images were automatically processed using the FreeSurfer software suite (surfer.nmr.mgh.harvard.edu), which was used to reconstruct the pial surface (GM/CSF boundary) and the white matter surface (WM/GM interface). Diffusion data was processed using the MRtrix3 toolkit (www.mrtrix.org). This step included Gibbs ringing removal, susceptibility distortion correction, eddy current correction, and bias field correction.

Network Architecture and Evaluation

To overcome the resolution constraints of clinical dMRI, we modified the DisC-Diff model to reconstruct high-resolution dMRI volumes. As illustrated in Fig. 1a, the model operated as a generative reverse diffusion process, learning to iteratively recover high-resolution (HR) b0 and b1000 images from Gaussian noise over 100 discrete timesteps. To adapt the framework for multi-modal guidance, we modified the core architecture to incorporate a specialized multi-stream encoding strategy (Fig. 1b). In addition, we introduced a dedicated parallel encoder specifically for the high-resolution T1-weighted anatomical reference. This design allows the model to explicitly disentangle anatomical priors from the diffusion signal streams. Furthermore, to improve feature integration, Squeeze-and-Excitation (SE) modules (Fig. 1c) were embedded within the network. These modules dynamically adjusted channel-wise

feature weights to balance the influence of anatomical priors (T1) and diffusion information during reconstruction. The model was trained and validated using high-quality data from the public Human Connectome Project (HCP) dataset.

To validate the reconstruction fidelity of DisC-Diff, all models were trained and tested under identical conditions to ensure a rigorous evaluation. We compared our proposed DisC-Diff framework with three distinct baseline methods representing different levels of super-resolution capability: Bicubic Interpolation, which served as a standard deterministic baseline; SR3 (Image Super-Resolution via Iterative Refinement), an advanced diffusion-based framework adapted for medical imaging; and a standard Denoising Diffusion Probabilistic Model (DDPM), representing a canonical generative approach without the specific disentanglement mechanisms incorporated in our model.

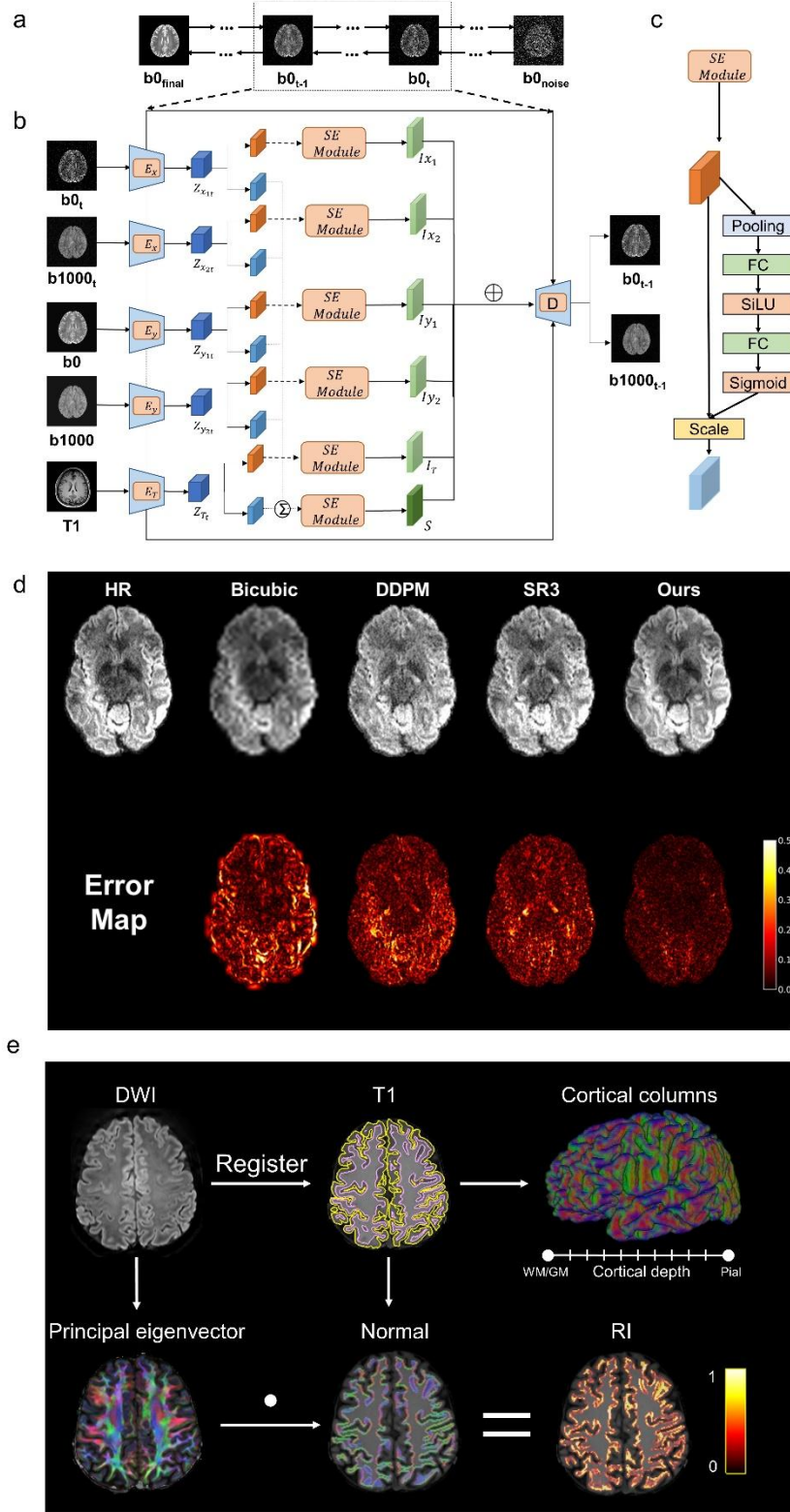


Fig.1: Disentangled Conditional Diffusion model (DisC-Diff) for super-resolution and column-based cortical analysis. (a) Conceptual workflow of DisC-Diff for dMRI super-resolution. In the forward diffusion process, Gaussian noise is progressively added to high-resolution (HR) dMRI, creating perturbed data. The backward diffusion process then denoises

this data, conditioned on the corresponding low-resolution (LR) dMRI. (b) The disentangled network contains five encoders to estimate the intermediate noise level. (c) The architecture of SE Modules dynamically weights disentangled representations. (d) Visual validation of super-resolution performance. Top row: Comparison of reconstruction fidelity between Ground Truth (HR), Bicubic interpolation, standard DDPM, SR3, and Ours (DisC-Diff) method. Bottom row: Corresponding voxel-wise error maps. (e) Column-based cortical microstructure analysis pipeline. Super-resolved DWI and T1 images are co-registered. T1-derived pial and WM/GM surfaces define the cortical columns. The RI is computed voxel-wise as the dot product between the principal diffusion eigenvector and the local cortical surface normal vectors, yielding a high-resolution cortical RI map.

Cortical Column-based Microstructure Analysis

As depicted in Fig. 1e, we implemented a column-based analysis using FreeSurfer-generated pial and white matter surfaces. Cortical columns were defined by linking topologically corresponding vertices between these surfaces. We discretized each column into 11 equidistant depth points. By co-registering super-resolved dMRI data, we computed the RI at each point as the dot product between the local diffusion eigenvector (V1) and the surface normal vectors. Based on the FreeSurfer automated segmentation (Desikan-Killiany atlas), the cerebral cortex was segmented into 74 distinct anatomical regions of interest (ROIs). To systematically parameterize the laminar organization, we extracted two key quantitative metrics from the depth-wise distributions: RI Skewness, quantifying the asymmetry of the laminar profile, and RI Maximum, representing the peak amplitude of radial integrity preserved along the cortical column. To characterize disease progression, we compared these metrics between groups (AD, MCI, HC) using independent t-tests. All results were corrected for multiple comparisons using the False Discovery Rate (FDR, $q < 0.05$).

Result

Validation of High-Fidelity Diffusion MRI Super-Resolution

Prior to clinical application, we validated DisC-Diff against high-resolution dMRI ($b = 1000$) ground truth (Fig. 1d). While standard interpolation (Bicubic) caused blurring and advanced generative models (SR3, DDPM) struggled with fine anatomical textures, DisC-Diff achieved superior reconstruction fidelity. The generated images were visually nearly indistinguishable from ground truth, a finding corroborated by voxel-wise error maps showing minimized residual errors compared to all competitors. By accurately recovering sub-voxel details without introducing artifacts, DisC-Diff established a robust, high-fidelity foundation for the subsequent column-level analysis of clinical ADNI data.

Laminar-specific Attenuation of Radial Index

To visualize the spatial and depth-dependent distribution of microstructural breakdown, we mapped the RI across three representative cortical depths: the pial surface, the mid-cortical layer (50% depth), and the WM/GM interface (Fig. 2a). In HC, the mid-cortical layers exhibited robust radial organization, characterized by high RI values (indicated by red and yellow intensities), which reflected the intact, vertically oriented column architecture, typical of the healthy cortex. However, this organization deteriorated across disease stages. As illustrated in Fig. 2a, the MCI group showed reduced high-RI regions, which further declined into a widespread attenuation of radial coherence in the AD group (predominantly yellow and cyan intensities). This observation was quantitatively confirmed by the depth-wise RI profile of relevant ROIs (Fig. 2b). The healthy cortex followed a typical inverted-U trajectory, peaking at approximately 50-60% of the cortical depth. In contrast, AD patients exhibited a significant flattened profile, with a reduction in peak amplitude across the Right Superior Parietal, Supramarginal, and Lateral Occipital cortices. The MCI group consistently occupied an intermediate position, preserving the profile shape but with reduced magnitude. The results supported the hypothesis that the loss of radial integrity is a continuous, stage-dependent process in AD.

Spatially Distinct Patterns of Laminar Profile Asymmetry and Attenuated Integrity

Vertex-wise statistical mapping revealed distinct spatial patterns of microstructural alteration across the cortex. As shown in Fig. 2c, comparisons between AD and HC (and AD vs. MCI) unveiled two specific patterns of microstructural breakdown. First, a widespread increase in RI Skewness (indicated by warm intensities) was observed, indicating a "laminar profile asymmetry". this asymmetry was predominantly located in posterior regions, affecting the visual and spatial attention networks, specifically the Lateral Occipital, Precuneus, and Superior Parietal cortices. Quantitative ROI analysis (Fig. 2d, top row) confirmed that AD patients exhibited highly significantly elevated skewness compared to both HC and MCI ($p < 0.0001$), with the MCI group often showing an intermediate deviation. In contrast, the reduction in RI Maximum (indicated by cool intensities) indicated an "attenuated peak radial integrity", which primarily affected the frontal executive networks. Significant regions of reduced peak integrity were identified in the Rostral Middle Frontal and Superior Frontal gyri, as well as the Inferior Parietal lobe. These regions (Fig. 2d, bottom row) showed a significant step-wise decline from HC to MCI to AD. Notably, the Inferior Parietal lobe, appeared to be a "confluence region", which exhibited both high skewness and low maximum, suggesting a compounding effect of profile asymmetry and integrity loss in this region.

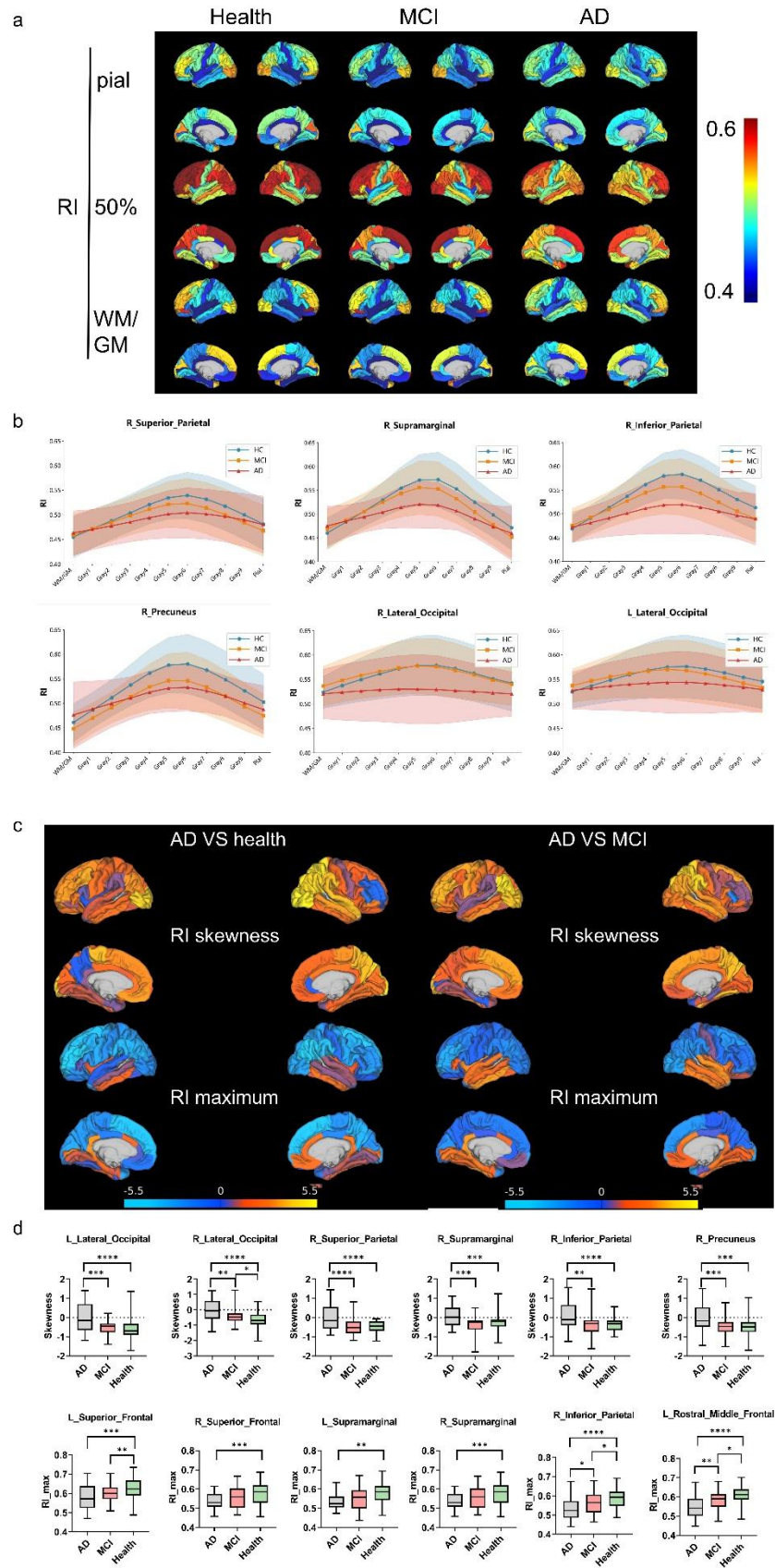


Fig.2: Laminar-specific attenuation and spatially distinct patterns of cortical microstructural alteration. (a) Cross-sectional visualization of RI maps across representative

cortical depths for Health, MCI, and AD groups. Red intensities indicate high RI, which is progressively attenuated in the MCI group and decreases to widespread low-RI values (orange/cyan) in the AD group. (b) Quantitative RI depth profiles for representative ROIs. Solid lines represent the group mean, and shaded areas indicate the standard deviation (± 1 SD). The healthy cortex (blue) displays an inverted-U trajectory, whereas AD patients (red) exhibit a flattened profile with reduction in peak amplitude. (c) Vertex-wise t-statistic maps comparing AD vs. Health (left) and AD vs. MCI (right). Top rows: high RI Skewness (warm colors) indicates laminar profile asymmetry, predominating in posterior occipito-parietal networks. Bottom rows: Reduced RI Maximum (cool colors) indicates attenuated peak radial integrity, localized primarily to frontal and parietal executive hubs. Color bars denote t-values (FDR-corrected). (d) Boxplots quantifying group differences in RI Skewness (top) and RI Maximum (bottom) within significant ROIs. Center lines represent medians; box limits indicate the 25th and 75th percentiles; whiskers extend to $1.5 \times$ the interquartile range. Statistical significance was determined using independent two-sample t-tests (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$).

Discussion

In this study, we applied generative AI super-resolution to overcome clinical imaging limits, providing an in vivo, depth-resolved map of cortical microstructure in AD. Our results challenged the view of uniform cortical decay. Instead, we revealed the spatial heterogeneity defined by two distinct microstructural signatures: a laminar profile asymmetry (high RI Skewness) predominating in the posterior occipito-parietal networks, and an attenuated peak radial integrity (low RI Maximum) mainly in frontal regions. Collectively, these spatially distinct signatures may reflect the underlying transition from early cytoarchitectural reorganization to macroscopic tissue loss, offering a more nuanced perspective on the propagation of AD pathology.

Cortical columns, vertical arrays of neurons facilitating intracortical communication, constituted the cytoarchitectural backbone of the cortex⁵. Histological evidence confirmed that the disruption of these arrays was an early biomarker of AD pathology. Diffusion MRI offered unique sensitivity to this organization via RI, which quantified vertical diffusion coherence. However, translating these insights into in vivo biomarkers has been impeded by the limited resolution of clinical dMRI (~2 mm), which needed the use of "depth-averaged" metrics. By averaged the signal across the cortical depth, such approaches obscured laminar-specific heterogeneity. Therefore, they failed to distinguish between uniform degeneration and complex, depth-dependent mechanisms, the laminar profile asymmetry and attenuated peak radial integrity. To address resolution limits without ultra-high-field hardware, we purposed the DisC-Diff model for diffusion MRI. While deep learning super-resolution was established, early CNN and GAN-based approaches often suffered from structural hallucinations or texture loss. Leveraging the advanced capabilities of DDPMs, we extended DisC-Diff, originally designed for structural MRI, to the complex domain of diffusion imaging. We demonstrated that DisC-Diff was uniquely suited for this task. By explicitly disentangling high-resolution anatomical priors (derived from T1) from the diffusion signal, our framework mitigated the hallucination risks inherent to standard GANs. Validated against previous methods like SR3, our approach yielded superior fidelity in

preserving fine-grained anatomical boundaries and tensor directionality. This computational breakthrough enabled the reliable reconstruction of cortical columns from standard clinical scans, unlocking microstructural insights previously inaccessible to conventional imaging.

Using high-resolution cortical columns, we identified two distinct microstructural signatures. First, the laminar profile asymmetry (high RI Skewness) in posterior visual-spatial networks, particularly the right occipito-parietal cortex, offered an important insight. While macroscopic atrophy studies often classified the occipital lobe as "spared" in typical AD, pathology studies confirmed it as a significant repository for A β deposition. Our findings suggested that profile asymmetry may serve as a sensitive indicator of this early-stage cytoarchitectural variation, detectable before volume loss. In addition, the frontal regions exhibited an attenuated peak radial integrity (low RI Maximum). This suggested that in frontal regions, the disease may have progressed to neurodegeneration, where the actual loss of neural substrate (cell death) flattened the radial column profile.

Reference

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